FLAb: Benchmarking deep learning methods for antibody fitness prediction [1]

Michael Chungyoun, [19]
Jeffrey Ruffolo,
Jeffrey Gray

NEURAL INFORMATION PROCESSING SYSTEMS
JOHNS HOPKINS UNIVERSITY

1 Background [2]

Antibody therapeutics must have a particular set of fitness properties [3]

a) Expression b) Thermostability c) Immunogenicity d) Binding affinity e) Aggregation f) Polyreactivity [4]

[5]

Six classes of biophysical data relevant to antibody development are (a) expression, (b) thermostability, (c) immunogenicity, (d) binding affinity, (e) aggregation, and (f) polyreactivity [6]

Do deep learning methods capture properties of therapeutic antibodies? [7]

We seek to answer this question by: [8]
1. Creating a collection of datasets of antibody therapeutic data
2. Benchmark representative AI model likelihoods for fitness prediction

2 Dataset curation [9]

We curated publicly available antibody fitness datasets [10]

Number of datapoints [11]

Antibody	Exp. (μg/mL)	T_m (°C)	Imm. (% ADA)	Binding (nM)	Agg. (Wv shift)	Poly. (min)
GSK CA1	34	34	-	29	-	-
GSK CA2	25	22	-	22	-	-
GSK CA3	11	8	-	11	-	-
GSK CA4	24	24	-	19	-	-
Hie C143	-	2	-	16	-	-
Hie mAb114	-	7	-	20	-	-
Hie mAb114 UCA	-	2	-	-	-	-
Hie MEDI8852	-	2	-	15	-	-
Hie MEDI8852 UCA	-	6	-	20	-	-
Hie REGN10987	-	8	-	13	-	-
Hie S309	-	10	-	19	-	-
Koenig G6	4275	-	-	4275	-	-
Prihoda imm	-	-	217	-	-	-
Rosace Adalimumab	-	14	-	14	-	14
Rosace CD3022	-	6	-	6	-	6
Rosace Golimumab	-	5	-	5	-	5
Shane. Trast. multi	-	-	-	24	-	-
Shane. Trast. zero	-	-	-	422	-	-
Warszawski D44	-	-	-	2049	-	-
Wittrup CST	274	137	-	-	822	411

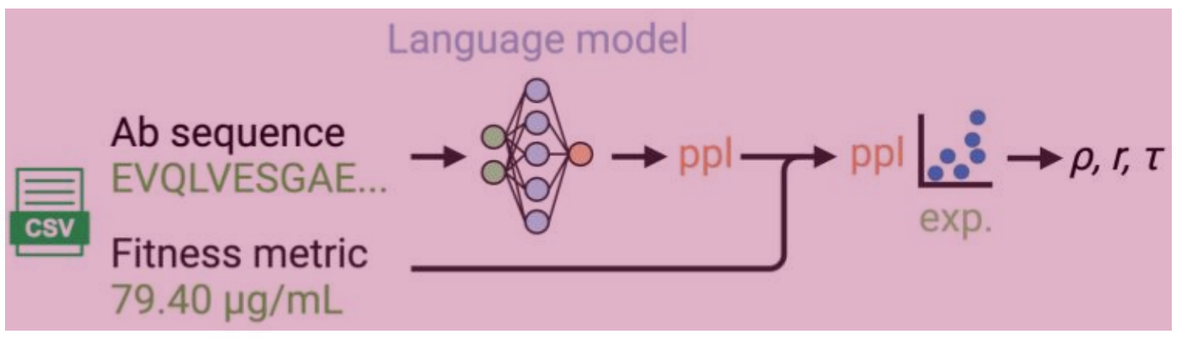
Dataset [13]

[12]

This database includes mutational landscapes of 17 distinct antibody families and a total of 13,384 associated fitness metrics. Each sequence is mapped to at least one fitness label pertaining to the 6 fitness properties [14]

3 Methods [15]

Pipeline for benchmarking protein language models [16]

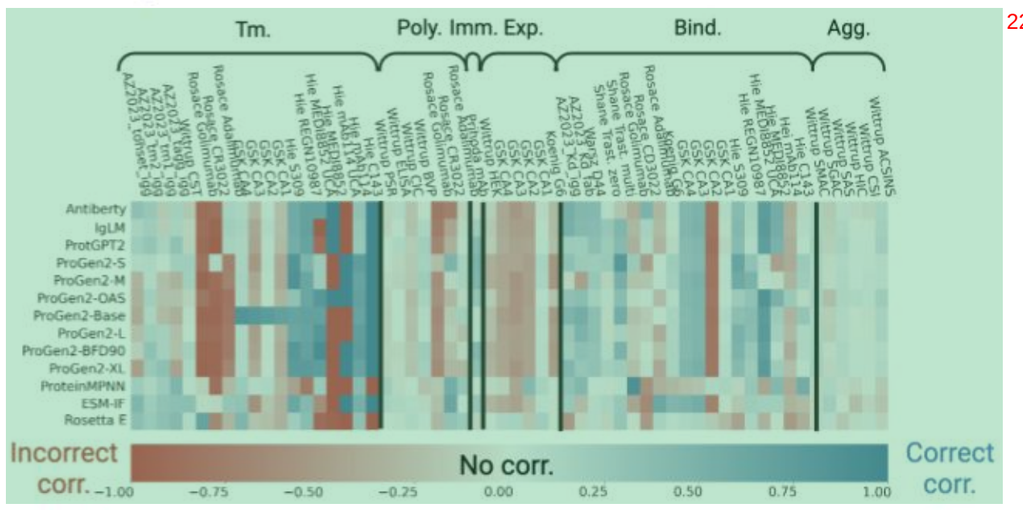


[17]

For each protein language model, we separately input the antibody **heavy** and **light sequences** to return two **perplexity scores**, and we tabulate the average perplexity between the two sequences. We calculate Spearman's (**ρ**), Pearson's (**r**) and Kendall tau's (**τ**) correlation coefficients between **perplexity** and **fitness**. [18]

4 Results [20]

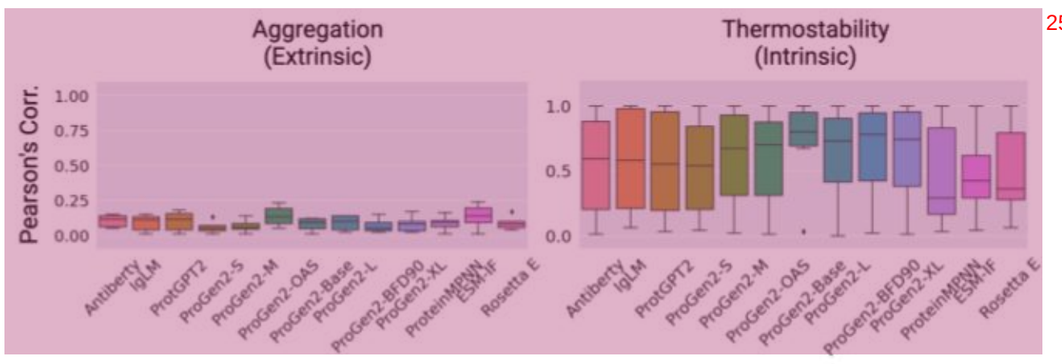
A heat map of model-dataset correlations [21]



[22]

Summary of Pearson's correlations. Models generally perform best with thermostability and binding affinity datasets, but most struggle with aggregation propensity and polyreactivity. [23]

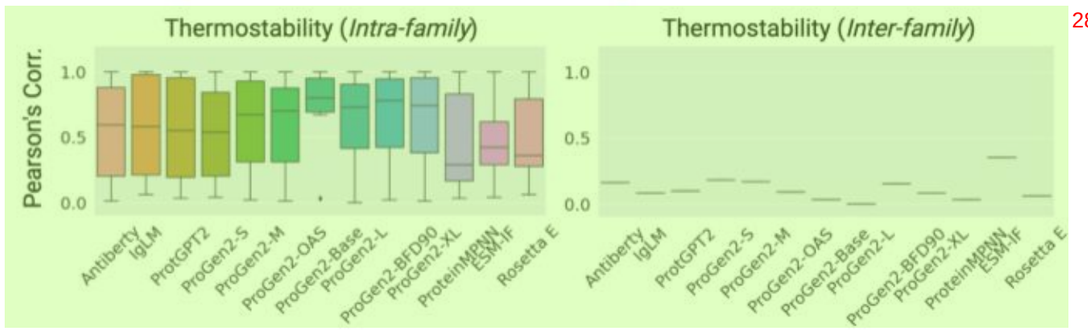
Intrinsic biophysical properties are more accurately predicted than extrinsic [24]



[25]

Intrinsic properties are impacted by inherent properties of the antibody. Extrinsic properties result from target biology and mechanisms of action. [26]

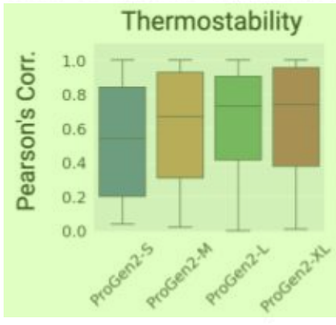
Models are more accurate at distinguishing intra-family versus inter-family antibody sets [27]



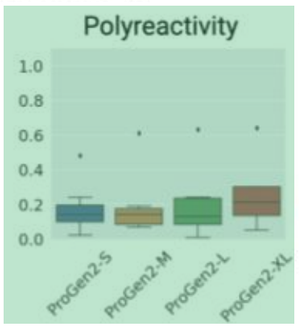
[28]

Intra-family antibodies are mutants originating from same wild type. Inter-family antibodies have different wild type origins. [29]

Parameter size influences performance over architecture and dataset composition [30]



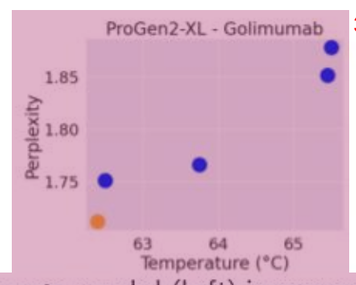
[31]



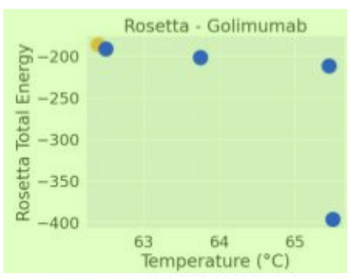
[32]

Polyreactivity and thermostability performance correlations improve with parameter size for ProGen2 (151M, 764M, 2.7B, 6.4B parameters) [33]

Some models favor evolutionary signal rather than physical fitness [34]



[35]



[36]

The language model (left) incorrectly assigns higher confidence to wild type antibody. Physics-based model (right) correctly assigns higher stability to the mutants [37]

5 Conclusion [38]

FLAb is a living benchmark where we will continuously add models evaluated and available antibody data [39]

No model was top performing across all six fitness classes [40]